Optimal Operation of the Electric, Natural Gas, and Water Systems

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PREFACE: This is the report of a study conducted by a research team at NC State University which aims at building an integrated framework for the operation of the electric, natural gas, and water systems, so as to improve efficiency, increase resilience, and ensure sustainability.

I. Introduction

Electric, natural gas, and water systems (EGWS) have been deeply coupled. The operation of some power plants and other energy production activities is constrained by water availability. The recent boom in natural gas further complicates the interdependence. Fig. 1 illustrates the interrelations between the electric, natural gas, and water system.

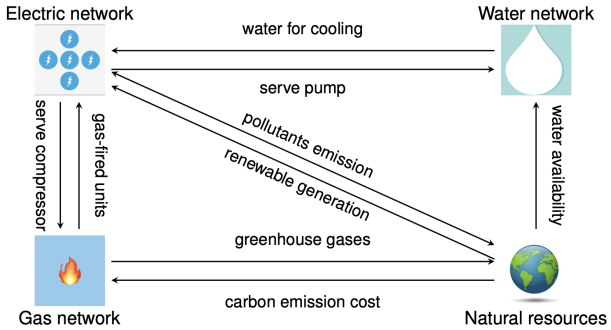

Fig. 1. The interrelations between electric, natural gas, and water systems. By leveraging the modern techniques in optimization and machine learning, the proposed project will build an analytical framework for the planning and operation of integrated energy-water systems.

In short-term operations, electricity generated by hydropower depends on the availability of streamflows. Moreover, the capacity of thermoelectric power plants relies on gas supply and cooling water [1]. Meanwhile, gas compressors need electricity to raise pressures in pipelines, and electricity is required to supply and treat water for agriculture, municipal, and industrial uses [2]. In gas production, water is needed for drilling, hydraulic fracturing, and well completion. Moreover, uncertainties in the EGWS are deeply intertwined. For example, forecasts of both daily streamflow and renewable generation are based on weather and climate inputs [3]. The realizations of water and gas demands depend on those of solar generation [4]. Conventional forecasting methods focus on the uncertainties in each individual system and ignore the correlations within the EGWS. From the long-term planning perspective, demands in all the systems are growing, while supply is constrained by limited resource availability and the impacts of climate change [5]. Under certain circumstances, utilities' plans for electricity capacity expansion could require much cooling water, which worsens summer water shortages in many regions of the country. A Warmer climate with higher carbon emission costs may lead to higher water demand for cooling purposes, and on the other hand, may stimulate the penetration of renewable energy which can potentially lower gas demand and prices. On the above premises, an integrated planning and operation model can make the EGWS more resilient, and ensure public well-being and sustainable growth.

Meanwhile, management of energy and water resources is influenced by a complex web of policies, practices, and institutions that collectively affect the evolution, planning, and operation of infrastructure at the nexus of energy-water systems [1]. Not surprisingly, the energy-water nexus (EWN) has been a subject of research, policy proposals, and more than a few implemented initiatives over the past few decades. Work to better understand and address the EWN ramped up markedly beginning with policy provisions in the 2005 Energy Policy Act [2] and subsequent work by the Government Accountability Office [3], Department of Energy [4], and Congressional Research Service [5], [6]. The emphasis in the literature followed a similar trajectory, with seminal works appearing in the late 2000s and early 2010s [7], [1] and followed by a rapid proliferation of diverse and expanded works [8], [9], [10].

Though inherently interrelated, multiple challenges exist to the integrated management of energy and water systems. The authors in [11] review several distinct but related barriers to integrated management, including: Traditionally independent and isolated sector management; Distinct spatial, temporal and physical characteristics; Data availability requirements; Degree of model aggregation and generalization; Complexity of multi-purpose reservoir topology and management; Collaboration of expertise and effort; Tracking changes in infrastructure and technological characteristics; and Uncertainty of energy and water futures. As alluded to in the first of these barriers, a central challenge to the integration of energy and water systems in the U.S. is the policy environment itself, including both specific policies and the broader institutions that feed into and support decision-making processes. There are broad regional differences in energy and water policy, and the locus of decision-making varies between each sector. Water governance varies both regionally and state-to-state, but generally falls into what can be described as prior appropriation, riparian,

and hybrid systems. Energy policy is strongly influenced by federal policy, but is likewise influenced by state-level and even local policy and resource considerations. Complicating matters even further are the dominant parties in each sector, with water infrastructure generally falling under the purview of public sector actors and energy infrastructure relying more on the private sector [6].

At present, energy and water governance are guided by a large number of specific laws and regulations. Oft-cited policy frameworks that govern the energy and/or water sector include the Clean Water Act, the Safe Drinking Water Act, the Clean Air Act, portions of EPAct 2005, the Energy Independence and Security Act of 2007, and the American Recovery and Reinvestment Act of 2009, various tax incentives, and numerous state and federal provisions governing the disclosure of fracking chemicals [12]. The wide array of the existing policy governing the energy and water sectors belies decision-making that is often narrowly focused and stove-piped within a given policy framework, with little consideration of impacts on resources outside of the direct control of deciding agency [3]. Despite the linkages between power and water infrastructure, it was not uncommon for decision-makers in one sector to simply assume that other resources will just be there to meet their specific needs [6].

II. Policy Options

In this section, we review research on the influence of discrete policies on energy-water systems planning and operation, from the perspective of policy options. Given the challenge of isolating specific outcomes from specific policy tools on specific technologies or practices [13], the review of research of policy options here follows the simplified categorization [5]. The review thus presents a brief discussion of research of policies relevant to the demand-side of each resource, policies relevant to the supply-side of each resource, and policies relevant to RD&D in each sector. An important variation is that the review below includes policies targeted to both water and energy sectors, as opposed to the energy sector-emphasis in [5].

A. Demand-Side Policy Options

Demand-side policy options are those which seek to reduce the demand for a particular resource. These include policies that primarily affect resource conservation and efficiency, whether through mandate or market incentive (e.g., tax, fee, or subsidy). Specific examples of demand-side policies could include technology standards that affect water consumption at an individual electric generating unit [8], [14], [13], reclamation or recycling requirements [15], [16], and/or resource pricing [8], [13], [17].

B. Supply-Side Policy Options

Supply-side policy options are those which seek to influence the manner in which a particular resource is provided. These include policies that affect the viability of a particular resource option relative to another, whether through mandate or market incentive (e.g., tax, fee, or subsidy). Specific examples of supply-side policies could include well spacing requirements [18], direct regulation of infrastructure operation or deployment decisions [14], [6], [19], [20], and/or pollution pricing such as a carbon price [8], [17].

C. RD&D Policy Options

Research, development, and demonstration (RD&D) policy options are those which seek to reduce resource impact by lowering the cost of alternative or more efficient uses of the resource. Also included in this category here are those policies which seek to increase the availability and quality of information to inform resource management decisions. Specific examples include direct RD&D funding as well as data collection, coordination, and quality control standards [6], [20], [13], [3].

III. Model Formulation

For the electric side, we consider a balanced distribution network integrated with distributed energy resources, including the battery energy storage system (BESS) and photovoltaic (PV) plants. Meanwhile, we consider a distribution-level gas system with gas compressors and gas storage systems. Gas-fired generators and power-to-gas stations connect the electric and natural gas systems. Such an integrated system is studied in the literature [21], [22]. In addition, a municipal water network is considered in our EGWS. The integrated system with electric and water network is presented in [23], [24]. In this section, we first model the operation constraints for the electric, natural gas, and water system separately and then formulate the EGWS optimal operation problem.

A. Modeling Power Distribution Networks

In this work, we consider a typical radial distribution network, and thus, the *DistFlow* model [25], [26] is adopted. Let the scheduling time horizon be T and $\mathcal{T} = \{1, \ldots, T\}$. Consider a distribution network as a directed graph $\mathcal{G} = (\mathcal{I}, \mathcal{E})$ where $\mathcal{I} = (1, \ldots, I)$ represents the set of nodes and $\mathcal{E}$ denotes the set of distribution lines connecting the nodes in $\mathcal{I}$. We index nodes and edges by i and (i, j), respectively, where (i, j) indicates pointing from i to j. Without loss of generality, we index the root of the tree (i.e., the substation) by 0. Each node $i \in \mathcal{I} \setminus \{0\}$ has an unique ancestor node and a set of children nodes. We denote them by $\mathcal{A}_i$ and $\mathcal{C}_i$, respectively. For each bus $i \in \mathcal{I}$, let $V_i = |V_i| e^{\mathbf{i}\theta_i}$ be its complex voltage, and we define $v_i := |V_i|^2$ as its magnitude squared. Meanwhile, P_i^{L} and Q_i^{L} represent the active consumption and reactive consumption at bus i, respectively. Let $z_{ij} = r_{ij} + \mathbf{i} x_{ij}$ and $S_{ij} = P_{ij} + \mathbf{i} Q_{ij}$ be the complex impedance and branch power flow from bus i to bus j. Let L_{ij} be complex branch current, and $l_{ij} := |L_{ij}|^2$ be the magnitude of complex branch current squared. For some nodes, they are with renewable energy generation (or gas-fired/diesel generators), battery energy storage system, and water pumps. Without loss of generality, we define $P_i^{\mathrm{R}}/Q_i^{\mathrm{R}}, P_i^{\mathrm{ES}}/Q_i^{\mathrm{ES}}, P_i^{\mathrm{P}}/Q_i^{\mathrm{P}}, \forall i \in \mathcal{I}$, to represent the active/reactive generation of renewable, discharging rates of

storage, and the consumption of water pumps at node i. It is worth noting that we simply set the corresponding values to zero if the nodes are not equipped with the elements. Moreover, the lower and upper bounds of a variable Z are denoted by $\underline{Z}$ and $\bar{Z}$, respectively. The detailed model for power distribution networks is given as

$$\sum_{j\in\mathcal{C}_i} P_{ij,t} - \sum_{k\in\mathcal{A}_i}\left(P_{ki,t} - r_{ki}l_{ki,t}\right) = P^{\mathrm{G}}_{i,t} + P^{\mathrm{R}}_{i,t} + P^{\mathrm{ES}}_{i,t} - PT_{i,t} - P^{\mathrm{L}}_{i,t} - P^{\mathrm{P}}_{i,t},\ \forall i\in\mathcal{I},\ \forall t, \tag{1a}$$

$$\sum_{j\in\mathcal{C}_i} Q_{ij,t} - \sum_{k\in\mathcal{A}_i}\left(Q_{ki,t} - x_{ki}l_{ki,t}\right) = Q^{\mathrm{G}}_{i,t} + Q^{\mathrm{R}}_{i,t} + Q^{\mathrm{ES}}_{i,t} - QT_{i,t} - Q^{\mathrm{L}}_{i,t} - Q^{\mathrm{P}}_{i,t},\ \forall i\in\mathcal{I},\ \forall t, \tag{1b}$$

$$v_{i,t} - v_{j,t} = 2r_{ij}P_{ij,t} + 2x_{ij}Q_{ij,t} - z^2_{ij}l_{ij,t}, \forall j\in\mathcal{C}_i, \forall i,t, \tag{1c}$$

$$P^2_{ij,t} + Q^2_{ij,t} = v_{i,t}l_{ij,t}, \forall j\in\mathcal{C}_i, \forall i,t, \tag{1d}$$

$$P^2_{ij,t} + Q^2_{ij,t} \le \bar{S}^2_{ij}, \forall (i,j)\in\mathcal{E}, \forall t, \tag{1e}$$

$$0 \le l_{ij,t} \le \bar{l}_{ij}, \forall (i,j)\in\mathcal{E}, \forall t, \tag{1f}$$

$$\underline{v}_i \le v_{i,t} \le \bar{v}_i, \forall t, i. \tag{1g}$$

It is worth noting that $P^{\mathrm{G}}_{i,t}$ is the total power generation at bus i and $PT_{i,t}$ is the power consumption of power-to-gas (P2G).

The nonlinear model of the BESS is incorporated into our model [27]. For all $i\in\mathcal{I}$ and $t\in\mathcal{T}$, we have

$$(r^{\mathrm{Batt}}_i + r^{\mathrm{Cvt}}_i)(P^{\mathrm{ES}}_{i,t})^2 + r^{\mathrm{Cvt}}_i(Q^{\mathrm{ES}}_{i,t})^2 = L^{\mathrm{ES}}_{i,t}v_{i,t}, \tag{2a}$$

$$(P^{\mathrm{ES}}_{i,t})^2 + (Q^{\mathrm{ES}}_{i,t})^2 \le (\bar{S}^{\mathrm{ES}}_i)^2, \tag{2b}$$

$$\underline{E}^{\mathrm{ES}}_i \le E^{\mathrm{ES}}_{i,0} - \sum_{\tau=0}^{t}(P^{\mathrm{ES}}_{\tau,t} + L^{\mathrm{ES}}_{\tau,t}) \le \bar{E}^{\mathrm{ES}}_i, \tag{2c}$$

where $P^{\mathrm{ES}}_{i,t}$ is the discharging rate of the BESS (charging rate if it is negative), r^{Batt}_i is the loss coefficient related to the battery, r^{Cvt}_i is the loss coefficient related to the converter. $P^{\mathrm{ES}}_{i,t}$ and $Q^{\mathrm{ES}}_{i,t}$ are the active and reactive power the the BESS at bus i, respectively. $L^{\mathrm{ES}}_{i,t}$ is the active power loss of the BESS at bus i, which is also a decision variable.

B. Modeling Gas System Networks

We use the steady state model to describe constraints in gas system. The gas flow $GL^{\mathrm{S}}_{m,t}$, in mille (1000) cubic feet (MCF) per hour (MCF/h), from gas supply at node m is constrained by physical characteristics, i.e. capacity limits of gas supplies $GL^{\mathrm{S,max}}_m$, which leads to

$$0 \le GL^{\mathrm{S}}_{m,t} \le GL^{\mathrm{S,max}}_m,\ \forall m\in\mathcal{M},\ \forall t. \tag{3}$$

Moreover, the pressure at each node needs to be bounded by the minimum pressure $\omega^{\min}_m$ and the maximum pressure $\omega^{\max}_m$, which is given by

$$\omega^{\min}_m \le \omega_{m,t} \le \omega^{\max}_m,\ \forall m\in\mathcal{M},\ \forall t. \tag{4}$$

We use Weymouth gas flow equations [28] to represent the gas flow through each pipeline. Again, we regard $s(p)$ and $r(p)$ as the sending and receiving nodes of pipeline p with pressure $\omega_{s(p),t}$ and $\omega_{r(p),t}$, respectively. The gas flow $GL_{p,t}$ of pipeline p in MCF/h, in time slot t, is modeled by

$$GL_{p,t}|GL_{p,t}| = \phi_p\left(\omega^2_{s(p),t} - \omega^2_{r(p),t}\right),\quad \forall p\in\mathcal{P}, \forall t, \tag{5a}$$

$$-GL^{\max}_p \le GL_{p,t} \le GL^{\max}_p,\ \forall p\in\mathcal{P}, \forall t, \tag{5b}$$

where ϕ_p and $GL^{\max}_p$ are a constant and the maximum capacity for pipeline p, respectively. $\mathcal{P}$ is the set of pipelines.

Let $\mathcal{G}(m)$ denote the set of gas units that belong to node m. The nodal balance in gas systems describes that the gas injection equals the gas withdrawn including the compressor consumption τ_m at each node m, which is given by

$$GL^{\mathrm{S}}_{m,t} + GT_{m,t} - \sum_{s(p)=m} GL_{p,t} + \sum_{r(p)=m} GL_{p,t} + GL^{\mathrm{dis}}_{m,t} - GL^{\mathrm{ch}}_{m,t} = \tau_{m,t} + D^{\mathrm{gas}}_{m,t} + GL_{g,t},\quad \forall m\in\mathcal{M},\ \forall p\in\mathcal{P},\ \forall g\in\mathcal{G}(m), \tag{6}$$

where $D^{\mathrm{gas}}_{m,t}$, $GT_{m,t}$, and $GL_{g,t}$ are the gas demand, total gas injected by P2Gs, and gas consumed by gas units in MCF/h, respectively, at node m in time slot t. $GL^{\mathrm{dis}}_{m,t}$ and $GL^{\mathrm{ch}}_{m,t}$ are the discharging and charging rates of the gas storage, at node m in time slot t.

The state of charge at time slot t depends on the state of charge of the previous time slot $t-1$. Let the maximum state of the charge, efficiency, and discharging and charging rates of storage at bus m be $SOC^{\max}_m$, η, $GL^{\mathrm{dis,max}}_m$, and $GL^{\mathrm{ch,max}}_m$. The model for gas storage is given by

$$0 \le GL^{\mathrm{dis}}_{m,t} \le GL^{\mathrm{dis,max}}_m,\ \forall m\in\mathcal{M},\ \forall t, \tag{7a}$$

$$0 \le GL^{\mathrm{ch}}_{m,t} \le GL^{\mathrm{ch,max}}_m,\ \forall m\in\mathcal{M},\ \forall t, \tag{7b}$$

$$SOC_{m,t} = SOC_{m,t-1} + \eta GL^{\mathrm{ch}}_{m,t} - GL^{\mathrm{dis}}_{m,t}/\eta,\quad \forall m\in\mathcal{M},\ \forall t, \tag{7c}$$

$$0 \le SOC_{m,t} \le SOC^{\max}_m,\ \forall m\in\mathcal{M},\ \forall t, \tag{7d}$$

where $SOC_{m,t}$ is the state of charge of storage in MCF, at bus m in time slot t.

The compressor is used to increase the gas flow pressure at the cost of gas consumption. The compression ratio is constrained by the maximum compression ratio R. The constraints of the gas compressor are given as follows:

$$\omega^{\mathrm{in}}_{m,t} \le \omega^{\mathrm{out}}_{m,t} \le R_m\omega^{\mathrm{out}}_{m,t},\ \forall m\in\mathcal{M},\ \forall t, \tag{8a}$$

$$H_{m,t} = E_m GL^{\mathrm{C}}_{m,t}\left[\left(\frac{\omega^{\mathrm{out}}_{m,t}}{\omega^{\mathrm{in}}_{m,t}}\right)^{0.236} - 1\right],\quad \forall m\in\mathcal{M},\ \forall t, \tag{8b}$$

$$\tau_{m,t} = \alpha_m + \beta_m H_{m,t} + \gamma_m H^2_{m,t},\quad \forall m\in\mathcal{M},\ \forall t, \tag{8c}$$

where $H_{m,t}$, $GL^{\mathrm{C}}_{m,t}$, and $\tau_{m,t}$ are heat rate, gas flow through the compressor, and gas consumption of the compressor at bus m in time slot t. E, α, β, γ are four parameters of the compressor. $\omega^{\mathrm{out}}_{m,t}$ and $\omega^{\mathrm{in}}_{m,t}$ are pressures at outlet and inlet of the compressor.

C. Modeling Water Networks

We make the following assumptions for the municipal water network. Most formulations are adopted from [23], [24].

1) The pipe network is a directed graph $\mathcal{G}_W = (\mathcal{K}, \mathcal{E}_W)$ where $\mathcal{K} = (1, \ldots, K)$ represents the set of junctions (nodes) and $\mathcal{E}_W$ denotes the set of pipes connecting the junctions in $\mathcal{K}$. We simply use $\mathcal{E}_W(k)$ to denote the neighbors of junction k and $\mathcal{E}_W^{\mathrm{P}}$ to denote the edges with pumps.
2) The pump is considered as a type of pipe that imposes a head gain when the pump is on.
3) The pump converts the electric power into mechanical power at a constant efficiency of η.
4) The power factors of pumps are fixed, namely, $P_{k,t}^{\mathrm{P}}/Q_{k,t}^{\mathrm{P}}$ is a constant.

Let incidence matrix be A such that $A_{kn} \in \{-1, 0, 1\}$, the water flow in pipes be $f_{kn,t}$, the flow into tank at node k be $f_{k,t}^{\mathrm{T}}$, and the flow injected by sources be $f_{k,t}^{\mathrm{G}}$. The mass balance of the water network is given by:

$$\sum_{kn \in \mathcal{E}_W(k)} A_{kn} f_{kn,t} = f_{k,t}^{\mathrm{G}} - f_{k,t}^{\mathrm{T}} - d_{k,t}^{\mathrm{w}}, \forall k \in \mathcal{K}, t, \tag{9}$$

where $d_{k,t}^{\mathrm{w}}$ is the water demand. Let R_{kn}^{P} be head loss coefficient of pipe (k, n), h_k be the elevation at node k, and y_k be the water head at node k. The hydraulic characteristics of a normal (without pump) pipe is given be:

$$\begin{aligned} y_{k,t} - y_{n,t} + h_k - h_n = R_{kn}^{\mathrm{P}} \mathrm{sgn}(f_{kn,t}) f_{kn,t}^2, \\ \forall (k,n) \in \mathcal{E}_W \setminus \mathcal{E}_W^{\mathrm{P}}, \forall t. \end{aligned} \tag{10}$$

To denote if the pump is on or off, we need to define a binary variable α_{kn}. If the pump is off, we simply set $f_{kn,t} = 0$. However, if $a_{kn} = 1$, the hydraulic characteristics are needed.

$$f_{kn,t} = 0, \text{if } a_{kn,t} = 0, \tag{11a}$$

$$\begin{aligned} y_{k,t} - y_{n,t} + h_k - h_n + y_{kn,t}^{\mathrm{G}} = R_{kn}^{\mathrm{P}} f_{kn,t}^2, \\ \text{if } a_{kn,t} = 1, \forall (k,n) \in \mathcal{E}_W^{\mathrm{P}}, \forall t, \end{aligned} \tag{11b}$$

where $y_{kn,t}^{\mathrm{G}}$ is head gain imposed by the pump in pipe (k, n). The water in tank should be constrained by its maximum and minimum requirements:

$$\underline{S}_k^{\mathrm{w}} \le S_{k,0}^{\mathrm{w}} + \sum_{t=0}^{t} f_{k,t}^{\mathrm{T}} \le \bar{S}_k^{\mathrm{w}}, \forall k, t. \tag{12}$$

Then, the following constraints are the bounds for some variables.

$$\underline{f}_{kn} \le f_{kn,t} \le \bar{f}_{kn}, \forall (k,n), \forall t, \tag{13a}$$

$$\underline{f}_k^{\mathrm{G}} \le f_{k,t}^{\mathrm{G}} \le \bar{f}_k^{\mathrm{G}}, \forall k, t, \tag{13b}$$

$$\underline{f}_k^{\mathrm{T}} \le f_{k,t}^{\mathrm{T}} \le \bar{f}_k^{\mathrm{T}}, \forall k, t, \tag{13c}$$

where f_{kn} is the vector of water flow in pipes. For simplicity, we can consider pumps as constant-speed motors. The hydraulic characteristics of a constant-speed pump is generally approximated by a quadratic function of the water flow across the pump, i.e., $y^{\mathrm{G}} = a_2 f^2 + a_1 f + a_0$, in which the contribution of the nonlinear $a_2 f^2$ is very small. As a result, the head gain is represented by

$$y_{kn,t}^{\mathrm{G}} = a_{kn,1} f_{kn,t} + a_{0,kn}, \forall (k,n), \forall t. \tag{14}$$

If variable-speed pumps are considered, the head gain is a quadratic function of both the flow and the speed, and the speed should be constrained [24]. It adds another layer of non-convexity. The variable-speed pumps are not considered in this project.

D. Modeling the Coupling Elements

The power system and gas system are coupled with gas-fired generators and P2Gs. The model carries out the fuel cost of a gas-fired generator by the cost of the gas supply. We use the higher heating value of synthetic natural gas HHV_{SNG}, taken as 38.42 MJ/m^3, to denote the conversion between electric power flow and volumetric gas flow per hour, which leads to the two equations below:

$$\begin{aligned} GT_{a,t} = & \eta^{\mathrm{P2G}} PT_{a,t}/HHV_{\mathrm{SNG}}, \\ & \forall a \in \cup \mathcal{A}(m), \forall t, \end{aligned} \tag{15a}$$

$$\begin{aligned} P_{g,t} = & \eta^{\mathrm{G2P}} GL_{g,t} HHV_{\mathrm{SNG}}, \\ & \forall g \in \cup \mathcal{G}(m), \forall t, \end{aligned} \tag{15b}$$

where $\mathcal{A}(m)$ denotes the set of P2G units that belongs to node m and $\mathcal{G}(m)$ denotes the set of gas units (for gas-fired generators) that belong to node m.

The following constraints act as the mathematical link between the water and power systems:

$$\eta P_{i,t}^{\mathrm{P}} = f_{kn,t} y_{kn,t}^{\mathrm{G}} = a_{kn,1} f_{kn,t}^2 + a_{0,kn} f_{kn,t}. \tag{16}$$

E. Objectives

Our objective is to minimize the operational cost of the integrated system, including the power system and gas system. It is worth noting that the electricity consumption of the pump is reflected in the total power consumption and thus is not listed as a separate cost in the objective. Say, the fuel cost function of the generator be $C_i^{\mathrm{p}}(\cdot)$ and the cost of gas supply be $C_m^{\mathrm{g}}(\cdot)$. Then we have our objective formulated as follows:

$$\min \sum_{t \in \mathcal{T}} \left[\sum_{i \in \mathcal{I}} C_i^{\mathrm{p}}(P_{i,t}^{\mathrm{G}}) + \sum_{m \in \mathcal{M}} C_m^{\mathrm{g}}(GL_{m,t}^{\mathrm{S}}) \right], \tag{17a}$$

$$s.t.\ (1) - (16) \tag{17b}$$

where $C_i^{\mathrm{p}}(P_{i,t}^{\mathrm{G}}) = a_i (P_{i,t}^{\mathrm{G}})^2 + b_i P_{i,t}^{\mathrm{G}} + c_i$ is the quadratic cost function of the generator. $C_m^{\mathrm{g}}(GL_{m,t}^{\mathrm{S}}) = \pi_t^{\mathrm{gas}} GL_{m,t}^{\mathrm{S}}$, where π_t^{gas} is random variable, representing gas price for each thousand cubic feet (MCF) at time t.

This problem is a mixed-integer nonlinear (nonconvex) programming (MINLP), which is hard to solve directly. Next section, we simplify the constraints and convert this problem into a mixed-integer convex quadratically constrained quadratic programming problem.

IV. Convex Relaxation and Linearization

A. Convex Relaxation of Electric System Constraints

In the electric system, we have two non-convex constraints, i.e., (1d) and (2a). To convexify these two constraints, we adopt the convex hull relaxation method from [23], where you can find the detailed formulation and proof. Here, we only give the convex hull of (1d) and (2a) as follows.

$$P_{ij,t}^2 + Q_{ij,t}^2 \leq v_{i,t} l_{ij,t}, \forall j \in \mathcal{C}_i, \forall i, t, \tag{18a}$$
$$\bar{S}_{ij}^2 v_i + \underline{v}_i \bar{v}_i l_{i,j} \leq \bar{S}_{ij}^2 (\underline{v}_i + \bar{v}_i), \forall j \in \mathcal{C}_i, \forall i, t, \tag{18b}$$

$$(r_i^{\text{Batt}} + r_i^{\text{Cvt}})(P_{i,t}^{\text{ES}})^2 + r_i^{\text{Cvt}}(Q_{i,t}^{\text{ES}})^2 \leq L_{i,t}^{\text{ES}} v_{i,t}, \tag{19a}$$
$$r_i^{\text{Batt}}(Q_{i,t}^{\text{ES}})^2 + \underline{v}_i L_{i,t}^{\text{ES}} \leq (\bar{S}_i^{\text{ES}})^2 (r_i^{\text{Batt}} + r_i^{\text{Cvt}}), \tag{19b}$$
$$(\bar{S}_i^{\text{ES}})^2 v_i + \underline{v}_i \bar{v}_i L_{i,t}^{\text{ES}} \leq (\bar{S}_i^{\text{ES}})^2 (\underline{v}_i + \bar{v}_i). \tag{19c}$$

B. Linearization of Gas System Constraints

It is obvious that the gas flow equation (5a) is nonlinear. We use the incremental model to derive a piecewise linear approximation of gas flow equation [29]. Using the $\Pi = \omega^2$ to replace the pressure squared, we have the new gas flow equation as follows:

$$GL_{p,t}|GL_{p,t}| = \phi_p \left(\Pi_{s(p),t} - \Pi_{r(p),t} \right), \quad \forall p \in \mathcal{P}, \forall t. \tag{20}$$

Let $\mathcal{Q}$ be the set of segments that are applied to linearize the left term, we introduce continuous variable κ_q and binary variable y_q for each segment $q \in \mathcal{Q}$. Continuous variables represent the length of each segment. Binary variables force that if a segment is chosen, all the other segments to its left must be chosen. The gas flow GL is approximated as:

$$GL = GL_1 + \sum_{q=1}^{|Q|} \left(GL_{q+1} - GL_q \right) \kappa_q, \tag{21a}$$
$$\kappa_{q+1} \leq y_q, \ y_q \leq \kappa_q, \ \forall q \in \mathcal{Q} - 1, \tag{21b}$$
$$0 \leq \kappa_q \leq 1, \ \forall q \in \mathcal{Q}. \tag{21c}$$

As for the nonlinear terms in (8), we adopt the method in [30]. We fit the gas consumption $\tau_{m,t}$ into polynomials of variables $GL_{m,t}^{\text{C}}$, $\omega_{m,t}^{\text{out}}$ and $\omega_{m,t}^{\text{in}}$ as the following equation:

$$\begin{aligned} \tau_{m,t} &= \sigma_1 GL_{m,t}^{\text{C}} + \sigma_2 {\omega_{m,t}^{\text{out}}}^2 + \sigma_3 {\omega_{m,t}^{\text{in}}}^2 + \sigma_4 \\ &= \sigma_1 GL_{m,t}^{\text{C}} + \sigma_2 \Pi_{m,t}^{\text{out}} + \sigma_3 \Pi_{m,t}^{\text{in}} + \sigma_4. \end{aligned} \tag{22}$$

C. Convex Relaxation of Water System Constraints

The constraint (10) can be as a quasi-convex hull, given as follows:

$$y_{k,t} - y_{n,t} + h_k - h_n \begin{cases} \leq (2\sqrt{2} - 2) R_{kn}^{\text{P}} \bar{f}_{kn} f_{kn,t} + (3 - 2\sqrt{2}) R_{kn}^{\text{P}} \bar{f}_{kn}, \\ \geq (2\sqrt{2} - 2) R_{kn}^{\text{P}} \underline{f}_{kn} f_{kn,t} + (3 - 2\sqrt{2}) R_{kn}^{\text{P}} \underline{f}_{kn}, \\ \geq 2 R_{kn}^{\text{P}} \bar{f}_{kn} f_{kn,t} - R_{kn}^{\text{P}} \bar{f}_{kn}^2, \\ \leq 2 R_{kn}^{\text{P}} \underline{f}_{kn} f_{kn,t} - R_{kn}^{\text{P}} \underline{f}_{kn}^2. \end{cases} \tag{23}$$

The nonconvex constraint (11) contains a logic proposition. To eliminate the if expression, we use the big-M technique to rewrite constraint (11) as

$$R_{kn}^{\text{P}} f_{kn,t}^2 - Y_1 \leq 0, \tag{24a}$$
$$Y_2 - R_{kn}^{\text{P}} \bar{f}_{kn,t} f_{kn,t} \leq 0, \tag{24b}$$
$$0 \leq f_{kn,t} \leq M * \alpha, \tag{24c}$$

where M is a sufficient large number, $Y_1 = y_{k,t} - y_{n,t} + h_k - h_n + y_{kn,t}^{\text{G}} + M * (1 - \alpha)$ and $Y_2 = y_{k,t} - y_{n,t} + h_k - h_n + y_{kn,t}^{\text{G}} + M * (\alpha - 1)$.

Constraint (16) is a quadratic equation that can be considered as the intersection of a convex inequality and a concave inequality. According to [23], the convex hull constraint (16) is given by

$$\eta P_{i,t}^{\text{P}} \geq a_{kn} f_{kn,t}^2 + b_{kn} f_{kn,t}, \tag{25a}$$
$$\eta P_{i,t}^{\text{P}} \leq (a_{kn} \bar{f}_{kn,t} + b_{kn}) f_{kn,t}, \tag{25b}$$

where $f_{kn,t}$ is nonnegative since the direction of the pump flows is determined.

V. Case Study

A. Test System

We consider the IEEE 13 node test feeder for the electric system [31], an 11 node gas system from GasLib [32], and an 8 node municipal water system [23]. The topology of the electric system is illustrated in Fig. 2. The connection with the gas system and water system are also marked. It is worth noting that the gas-fired generator and the P2G plant connect the electric and the gas system, while the water pump is considered as a constant load when it is on. There are two PV plants deployed on buses #633 and #680, both with a 0.5 MW capacity rate. The forecast of the PV generation will be introduced later. Here, we use a fixed PV profile from [23]. Two identical battery energy storage systems are placed on buses #684 and #692. The capacity is 0.5MWh with a charge/discharge rate as 0.25 MW.

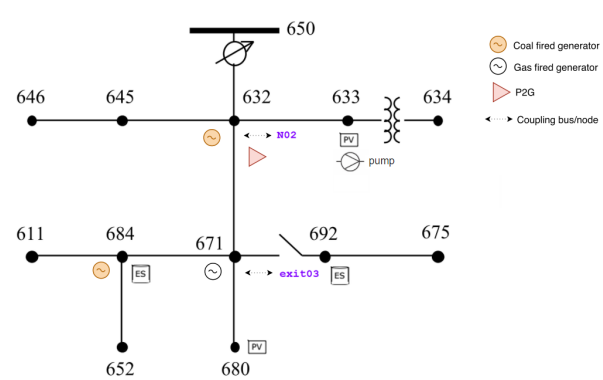

Fig. 2. Topology of the IEEE 13 node test feeder.

The structures of the gas system and the water system are given in Fig. 3 and Fig. 4. The gas system has 3 entries (gas supply nodes) and 3 exits (gas demand nodes), as well as two compressors used to increase the gas pressure. The water system has an unlimited water source, an electric pump, and a

water tank. When the pump is on, the water will flow into the nodes and the water tank, while the tank will serve the load when the pump is off. The detailed parameters of this water network can be found in [33].

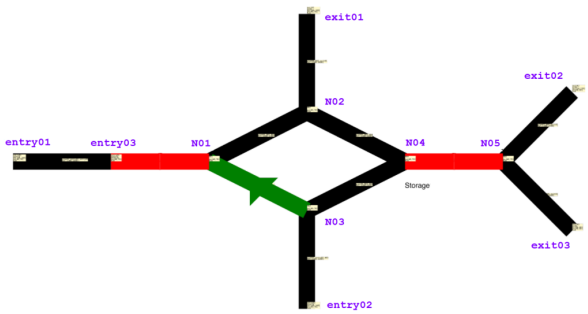

Fig. 3. Topology of a 11-node natural gas system. There are 3 gas supply nodes, 3 load nodes, and 2 compressors.

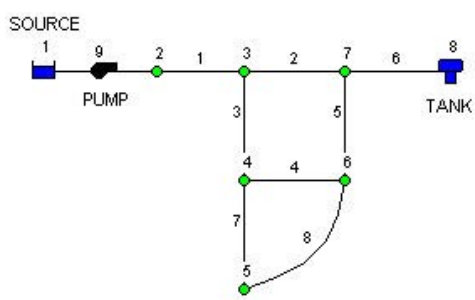

Fig. 4. Topology of a 8-node water network. There is 1 water source (unlimited), 1 pump, and 1 tank.

B. Results

We solve the above problem on a platform running a Linux distribution with kernel version 5.10.92-1 on an Intel Skylake CPU with 4 hardware threads, a maximum clock frequency of 2.30 GHz, and 3072 KB of level-1 cache per thread. The platform has around 15 GB of available DDR4 physical memory. The solver package was Gurobipy 9.5.0 build v9.5.0rc5 with an academic lincese[1]. The program was executed by Python interpreter version 3.9.2. The problem contained 2 quadratic objective terms, a total of 250 decision variables, of which 248 were continuous and 2 were binary, 260 linear constraints, and 68 quadratic constraints. To simplify the calculation, the scheduling horizon is set as 2 hours. The solution was obtained after 23 iterations in a total of 0.03 seconds.

C. Issues and Workarounds

1) Units of Measure: The model involved 3 different systems each having their individual convention for units of measure. Converting quantities into SI units in favor of uniformity were initially considered but dismissed during a discussion due to the following

- The datasets were primarily imperial.
- SI units could cause excessively small or large constants, resulting in an ill-conditioned matrix in the solver that may exacerbate the loss of precision in IEEE-754 based floating point arithmetics.

Workarounds: The chosen workaround was to select a list of units that creates constants with reasonable magnitudes and employ unit conversions extensively to ensure the meaningfulness of the results. An alternative workaround to address the unit-related numeric issue is to use an SI-based per-unit system for the electrical, gas, and water components, for which the suitable bases are selected such that the magnitude of the resulting quantities are around 1. This approach was not adopted since the details of the components were not initially clear during the early stages of implementation, and out of the fear that the per-unit system would further complicate the problem and obstruct the diagnosis of potential defects.

2) Big-M Method: The constant M used in constraints formulated using the big-M method resulted in an ill-conditioned matrix in the solver that may exacerbate the loss of precision in IEEE-754 floating-point arithmetics.

Workaround: The big-M method was only used for constraining the water flow to the on-off status of the pump. The constant M was selected such that it was larger than the all higher bounds of the water flow in pump-enabled pipelines, so that the flows are not affected by the arbitrary selection of M, but not large enough to excessively increase the condition number of the matrix in the solver.

3) Sampling Frequency: The original mathematical model was based on the assumption that the scheduling horizon and all-time series data had a uniform sampling rate of 1 (unit of time), since a number of equations implicitly multiplied by 1 (unit of time).

Workaround: A clause was added in the Contract of Implementation that designates the scheduling interval to be 1 hour and units of all quantities involving multiplication or division by time to be hour-based.

4) Fields of Definition of Symbol Sets: The fields of definition of symbol sets were unclear or unspecified.

Workaround: A Contract of Implementation was created to clarify the unspecified details in the model. The contract was periodically reviewed to ensure the state of coherency.

5) Feasibility of the Water Component: The water component, when isolated from the rest of the system, was infeasible. It was found that after the removal of a water pump on line 7-8, the water model contained insufficient degrees of freedom. The constraint set resulted in an empty feasibility space.

Workaround: The dataset for the water component was modified so that the topology was operative under a single pump.

VI. Load Forecasting

Electric, natural gas, and water systems have been deeply coupled. Conventional forecasting methods focus on the uncertainties in each individual system and ignore the correlations within the EGWS. In this section, we propose a multi-output forecasting model for concurrent forecasting of electricity,

[1]An academic license was used since the number of decision variables exceeded the limitation of the free license.

natural gas, and water demands. We will also show the forecast results using the independent models (load forecast models for individual systems) and compare the results.

Considering that the demand data for all three systems within the same geographical location is very difficult to acquire, we use a single household dataset with minute-level load data and additional temperature data for all three systems. The dataset has 730 days of captured data and has been pre-cleaned for providing consistent and comparable results [34]. The load data has been converted into an hourly resolution for the forecast. We also assume the considered topology to have a similar overall demand profile for all three systems. Since the available data is very limited, we try two different feature engineering approaches. In one approach, we use one-hot encoding for the categorical features such as the hour of the day, day of the week, the month of the year, etc., and use it for all the models. In the other approach, we label the categorical features and normalize them to be used with only the neural network-based models.

We include the interaction of hour of the day and temperature in our model to represent the fact that the demands of all three systems are affected by their respective demand of previous days. To make the model aware of recent changes in demand patterns, we also include autoregressive features to incorporate recency effects. For the sake of comparison between both the approaches, we include the same calendar and temperature parameters in both models but for the independent models based approach, we include autoregressive features corresponding to just each individual system whereas in the combined model approach, we include autoregressive features for all the three systems simultaneously. Using these two different approaches, we can also analyze the interdependence of each demand pattern for the prediction of all other demand patterns which can stand for typical human behavior for when people leave for work in the morning and come back in the evening.

We use independent models for all three systems and compare the performance with combined multi-output models developed using machine learning-based methods which can inherently support multi-output prediction such as linear regression, k-nearest neighbor, and random forest along with neural network-based models which can support multi-output predictions. The results achieved using all the different models are averaged over multiple runs and are given in Table I to Table IV. Both the root mean square error (RMSE) and mean absolute error (MAE) are used as the metrics for performance evaluation. Since all the demand patterns are correlated to each other, using the combined model or independent models for each individual system does not yield any significant difference in their performance.

TABLE I
RMSE OF COMBINED FORECAST MODELS

Model	Electricity	Natural Gas	Water
Linear Regression	628.45	194.57	0.64
K-Nearest Neighbor	689.18	229.14	0.73
Random Forest	882.16	273.76	0.88
Multilayer Perceptron (Approach 1)	853.22	248.42	1
CNN - LSTM (Approach 1)	757.04	271.79	0.82
Multilayer Perceptron (Approach 2)	877.3	266.02	0.87
CNN - LSTM (Approach 2)	755.08	243.14	0.8

TABLE II
RMSE OF INDEPENDENT FORECAST MODELS

Model	Electricity	Natural Gas	Water
Linear Regression	621.69	194.49	0.63
K-Nearest Neighbor	685.48	221.36	0.72
Random Forest	847.92	270.64	0.95
Multilayer Perceptron (Approach 1)	907.84	260.40	1.11
CNN - LSTM (Approach 1)	751.41	254.61	0.81
Multilayer Perceptron (Approach 2)	815.43	258.34	0.75
CNN - LSTM (Approach 2)	752.22	315.85	0.81

TABLE III
MAE OF COMBINED FORECAST MODELS

Model	Electricity	Natural Gas	Water
Linear Regression	462.62	137.16	0.41
K-Nearest Neighbor	481.25	162.52	0.46
Random Forest	591.27	195.4	0.53
Multilayer Perceptron (Approach 1)	611.38	172.28	0.69
CNN - LSTM (Approach 1)	574.37	176.36	0.64
Multilayer Perceptron (Approach 2)	588.45	176.56	0.6
CNN - LSTM (Approach 2)	562.06	176.31	0.59

TABLE IV
MAE OF INDEPENDENT FORECAST MODELS

Model	Electricity	Natural Gas	Water
Linear Regression	458.41	136.98	0.41
K-Nearest Neighbor	481.21	155.04	0.47
Random Forest	570.39	189.00	0.55
Multilayer Perceptron (Approach 1)	644.98	176.98	0.69
CNN - LSTM (Approach 1)	562.09	176.72	0.53
Multilayer Perceptron (Approach 2)	531.53	173.05	0.47
CNN - LSTM (Approach 2)	573.93	279.21	0.53

VII. CONCLUSION

Given the complexity of governance of the electrical, natural gas, and water systems (EGWS), systems integration remains a challenging undertaking in practice. In this project, we solve the problem of optimal operation of the EGWS and investigate the potential benefit of using combined models for load forecast of EGWS. By leveraging the convex relaxation techniques, we convert the original nonconvex nonlinear problem into a mixed-integer convex quadratically constrained quadratic programming problem, which can be solved efficiently using commercial solvers. Due to the nonconvexity of the original problem, we are not able to identify the global solution and thus cannot verify the solution gap. In terms of the combined forecast model, the results show that the forecast accuracy is not significantly better than the counterparts of the independent model. This is because all the demand patterns are correlated to each other, using the combined model or independent models for each individual system does not yield any significant difference in their performance.

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